

Brandon Higbee

Chapel Hill, NC • brandonhigbee95@hotmail.com • (318) 321-2455
<https://www.linkedin.com/in/brandon-higbee-557b5a139>

Summary

Senior Software Engineer with **8 years** of experience building scalable backend systems and full-stack web applications. Led development of **Python** and **SQL**-based data processing platforms serving enterprise clients across healthcare and SaaS domains. Designed high-performance **RESTful APIs** with authentication systems securing sensitive educational and business data. Managed cloud infrastructure on **AWS** while mentoring cross-functional engineering teams of 4-6 members. Seeking to apply **performance optimization, caching systems**, and distributed computing to complex backend platform challenges. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Programming Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS

Backend & APIs: FastAPI, Node.js, RESTful APIs, GraphQL, OAuth2

Databases: PostgreSQL, MongoDB, Redis, DuckDB

Infrastructure & DevOps: AWS, Docker, Kubernetes, CI/CD

Performance & Architecture: Caching systems, Graph-based dependency frameworks, Performance optimization, Systems architecture, Terraform

Work Experience

Senior Software Engineer | Tackle.io | 03/2022 - Present

- Architected backend data processing pipeline handling 2M+ daily transactions using **Python** and **SQL**, reducing query response times from 8s to under 1.5s through query optimization and indexing strategies.
- Built intelligent **caching systems** with **Redis** for frequently accessed data, achieving 40% reduction in database load and improving API response times by 3x.
- Led migration of monolithic backend services to microservices architecture using **Docker** and **Kubernetes**, improving deployment frequency from monthly to daily releases.
- Designed and implemented **RESTful APIs** with **OAuth2** authentication for SaaS platform integration, enabling 50+ enterprise partners to automate workflows securely.
- Collaborated with product and data teams to define requirements for **performance optimization** initiatives, prioritizing work based on customer impact metrics.
- Mentored 3 junior engineers through code reviews and pair programming sessions, fostering growth in **Python** backend development and system design.
- Established API design standards and documentation practices adopted across 12-engineer team, reducing onboarding time for new developers by 35%.
- Implemented **graph-based dependency framework** for optimizing multi-step execution plans, cutting processing time for complex workflows by 60%.

Software Engineer | Tackle.io | 06/2021 - 03/2022

- Created **GraphQL** API layer for complex data querying needs, enabling frontend teams to fetch nested data efficiently without multiple REST calls.
- Implemented background job processing system with **Redis** queues, handling 10,000+ async tasks daily for report generation and data synchronization.
- Developed **Python** backend services powering multi-tenant SaaS platform, supporting 200+ concurrent users with **PostgreSQL** database optimization.
- Participated in on-call rotation resolving production incidents, maintaining 99.9% platform uptime for critical customer-facing applications.

Software Engineer II | Ultimate Medical Academy | 04/2019 - 06/2021

- Built **RESTful APIs** with **Node.js** and **Express** for healthcare education platform serving 50,000+ students, implementing JWT-based authentication securing protected health information.
- Optimized **SQL** queries in **PostgreSQL** reporting database, reducing complex enrollment report generation from 45s to under 5s for administrative dashboards.
- Designed **MongoDB** schemas for course content management system, supporting hierarchical data structures and enabling rapid content iteration by editorial team.
- Integrated third-party learning management systems via API connections, automating student enrollment data synchronization and saving 20+ hours weekly in manual data entry.

Software Engineer I | Ultimate Medical Academy | 08/2018 - 04/2019

- Assisted in database migrations and schema updates, ensuring data integrity during platform feature releases.
- Contributed to backend API development with **Node.js**, implementing CRUD operations for user profiles and course progress tracking.
- Developed student-facing web features using **React** and **JavaScript**, improving course completion rates by 15% through enhanced UI interactions.

SDK Intern | Accusoft | 06/2017 - 08/2018

- Developed **JavaScript** SDK components for document processing library, writing unit tests achieving 85% code coverage.
- Created sample applications demonstrating SDK integration patterns, reducing customer support tickets by 25%.

Education

Bachelor of Science

University of South Florida | 08/2014 - 05/2018